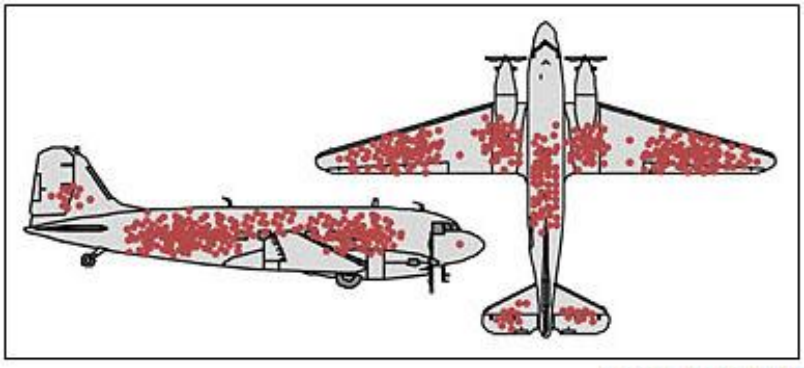
10. Design Thinking

"Design Thinking is a mindset, not a toolkit or a series of steps"

Ricardo Filipe, ISEC (ricardo.filipe@isec.pt)

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Motivation – think different



Credit: Cameron Moll

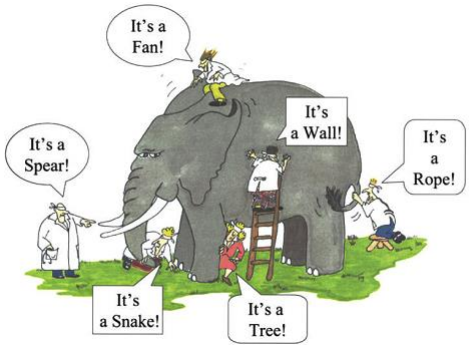
286

Motivation



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Motivation



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Practical Examples





https://www.ideou.com/blogs/inspiration/from-design-thinking-to-creative-confidence

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Practical Examples



https://www.reddit.com/r/mildlyinteresting/comments/bsyk29/this_retirement_home_has_a_fake_bus_stop_to_keep/

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Practical Examples

- AirBnb



Looking through the ads of the landlords, they realized that photos of all houses were pretty similar. Here are the features that caught the eye:
•the images are mostly taken on a smartphone
•they are of poor quality
•they show some rooms, but not all of them

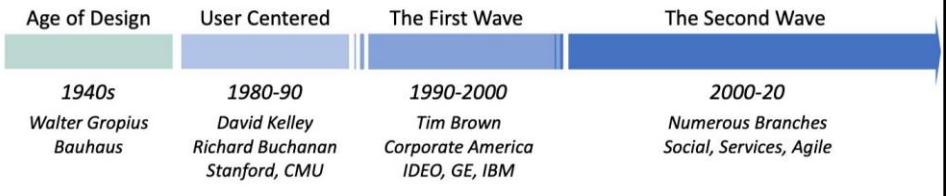
https://www.bbva.com/en/airbnb-design-thinking-success-story/

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History of Design Thinking

Design Thinking History



Richard Buchanan's paper, **"Wicked Problems in Design Thinking"** written in 1992 describes the roots of Design Thinking that go back to problem solving and innovation social science.

Design Thinking was then popularized by the Stanford professor, David Kelley, who was the founder of the d.school.

IDEO, which brought Design Thinking into the mainstream, was founded as a merger between David Kelley Design (Kelley designed Apple's first mouse) and another Industrial Design firm in 1991.

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What is Design Thinking?

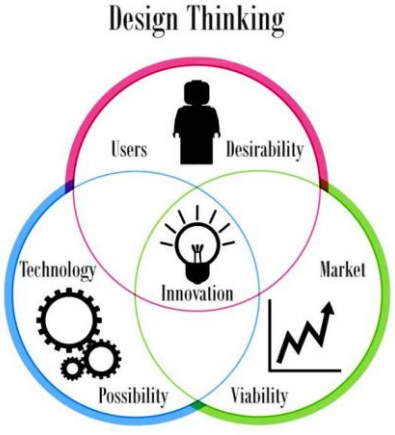
- Design thinking is **a process for solving problems by prioritizing the consumer's needs above all else**. It relies on observing, with empathy, how people interact with their environments, and employs an iterative, hands-on approach to creating innovative solutions.

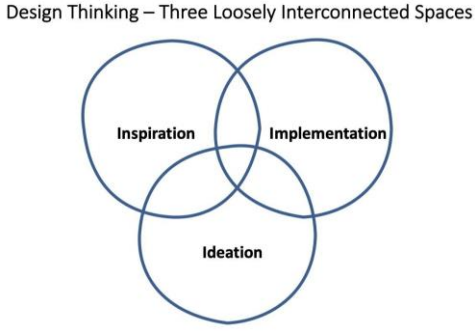
- *Design thinking can be described as a discipline that uses the designer's sensibility and methods to match people's needs with what is technologically **feasible** and what a viable **business** strategy can convert into customer value and market opportunity.*

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What is Design Thinking? – 3 lenses

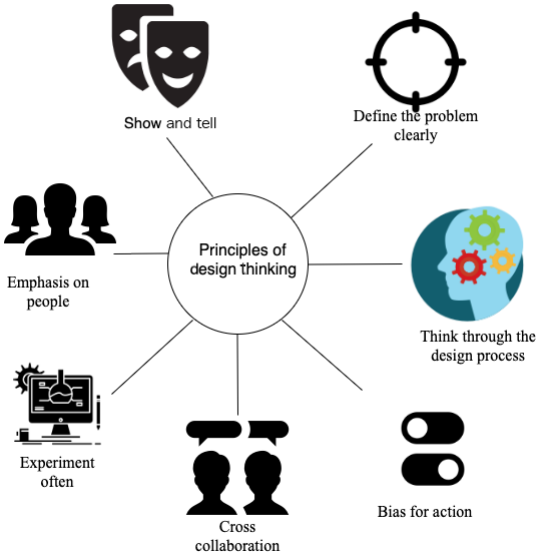




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What is Design Thinking? – Principles

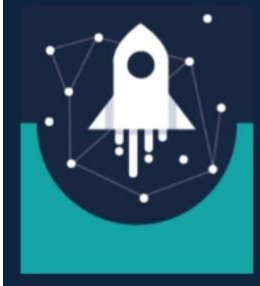


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What is Design Thinking? – Principles

Benefits of Design Thinking



- Early focus on understanding users --
  - Allows teams to agree on the mission early.
  - Provides a strong basis for later steps in the process.
- Experimental approach of presenting ideas as hypotheses –
  - Reduces the risk and cost of failure
  - Reduces cognitive biases
  - Allows for easier acceptance of failures

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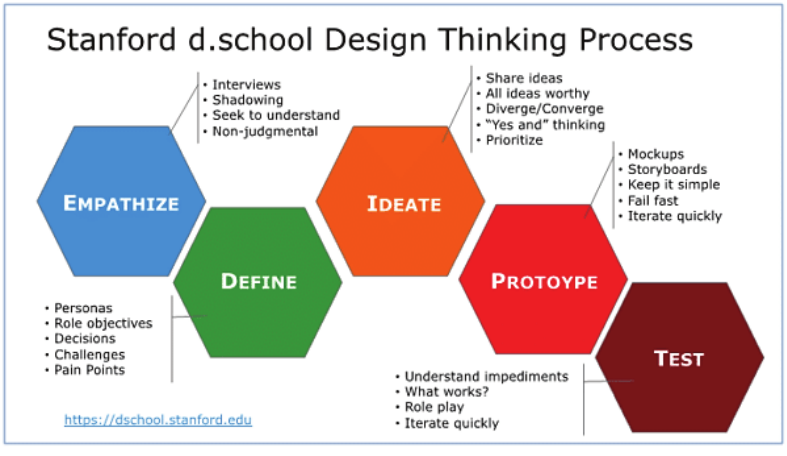
Why Design Thinking?

- Problem not defined
- Closer to the customer
- For new products and service development

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Design Thinking Steps

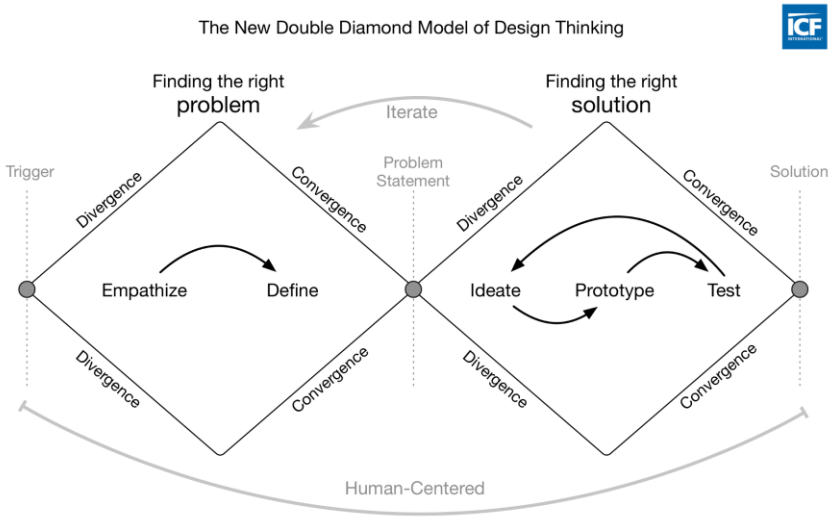


Understand the problem — Find a solution

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DS – Diverge and Converge Diamond



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Empathize

- Understand your customer pain.
- The Empathize mode is the work you do to understand people, within the context of your design challenge.
- It is your effort to understand the way they do things and why, their physical and emotional needs, how they think about world, and what is meaningful to them.

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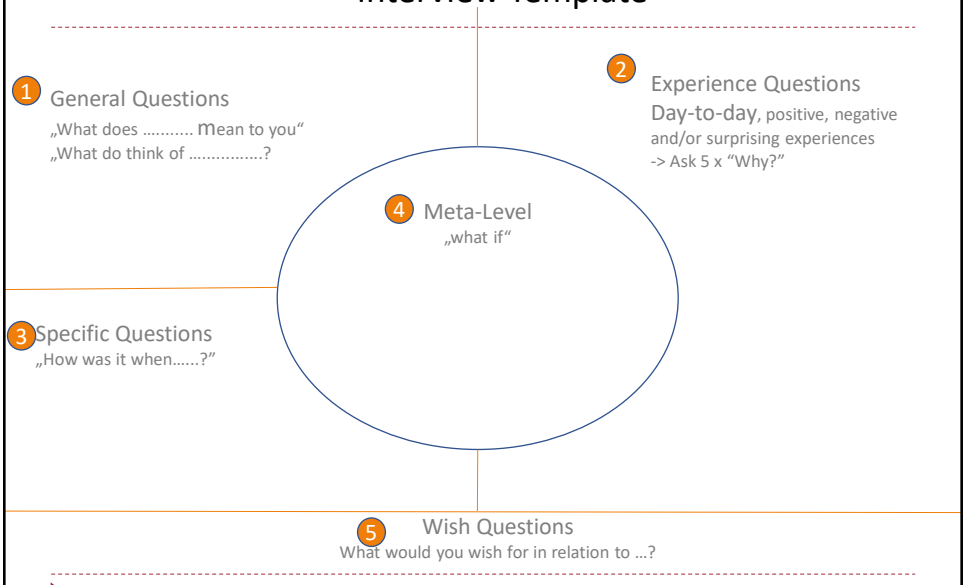
Empathize

- Observe
  - **Fly-on-the-wall**
  - View users and their behavior in the context of their lives. As much as possible do observations in relevant contexts in addition to interviews. Listen more than you talk.
- Engage
  - **Qualitative interviews**
  - Prepare some questions you'd like to ask, but expect to let the conversation deviate from them. Keep the conversation only loosely bounded. Elicit stories from the people you talk to, and always ask "Why?" to uncover deeper meaning.
- Watch and Listen
  - Self Immersion

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Interview Template



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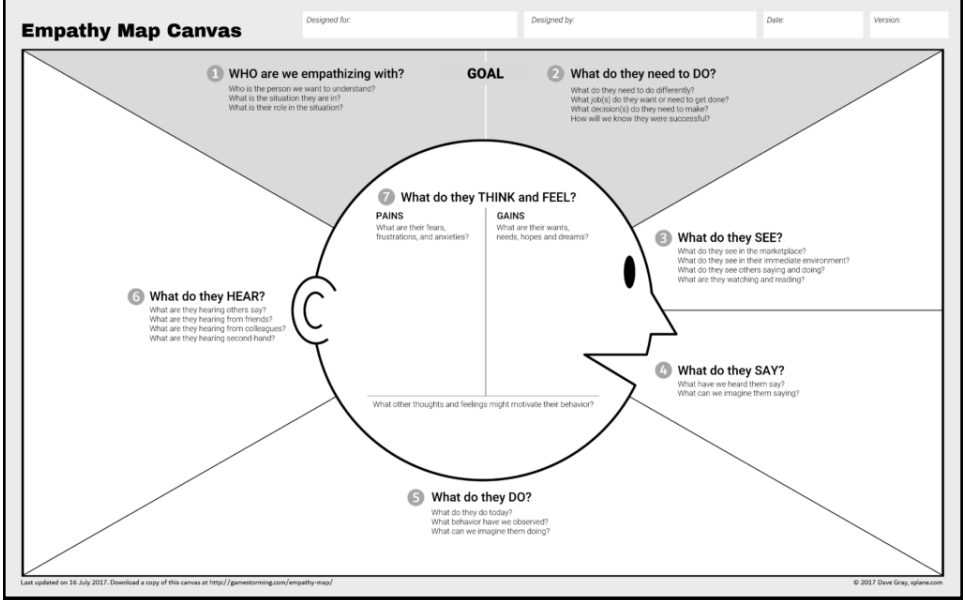
Define

- Create a persona
  - *A persona is a synthesized description of your target group*
  - *It's a combination of (research) knowledge and empathy*
  - *Creation of a target group*

- *Persona has:*
  - *A specific demographic*
  - *Desires*
  - *Needs*
  - *Problems*
  - *…*

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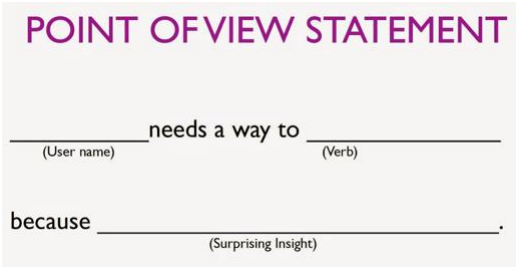
Define – techniques: empathy map



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Define

- Create a problem statement:
  - Orientation on the problem
  - Common understanding
  - Focus in the solution



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Ideate

- Using the knowledge of previous phases, create ideas to solve the problema.

- 2 phases:
  - Generate ideas
  - Sort and filter ideas

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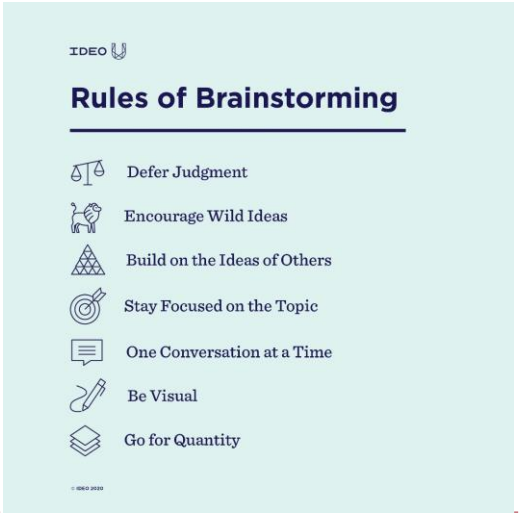
Ideate

- Generate Ideas
  - Braindump/Brainstorming/Brainwriting
  - Association
    - **word storm**
      - *A word storm is where you write down the words that come to mind when you see another word. You might start with a word or two based on your project or challenge, and begin writing down any word that comes to mind. These words are then grouped together according to how they are related to each other.*
    - **word association**
      - Word associations are the same as a word storm, except that you don't group according to how the words are related to each other. It works better as a technique to get creativity flowing when you don't want to bother with over-thinking how words are related.
    - **mind mapping**
      - *a mind map is a way to visually organize data and information.* Organized around a central idea, a mind map works like the branch of a tree. Ideas and then sub ideas that are associated with the main idea branch off from the central idea.
    - **visual association**

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Ideate

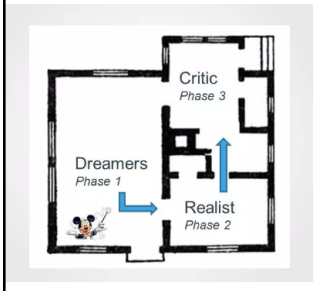


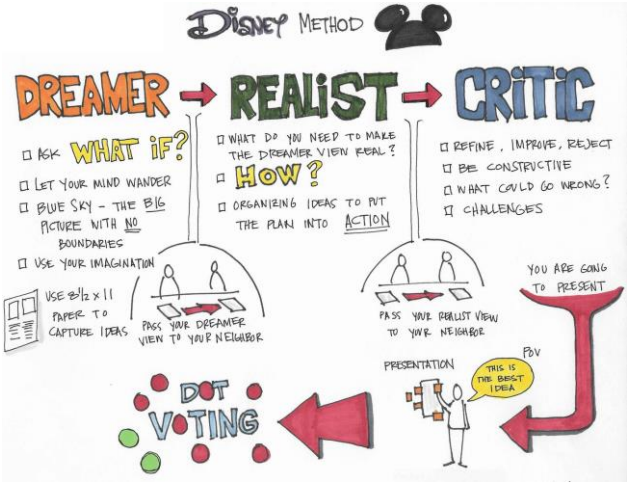
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Ideate

- Generate Ideas techniques:
  - Disney Method
    - 3 stages





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Ideate

- Generate Ideas
  - De Bono Thinking Hats
    - The Six Thinking Hats is a role-playing model developed by Edward de Bono in 1986. Each hat represents a different lens or perspective on a particular issue and is an insightful activity that prevents narrow thinking.
  - Generates ideas based on diferent prespectives.

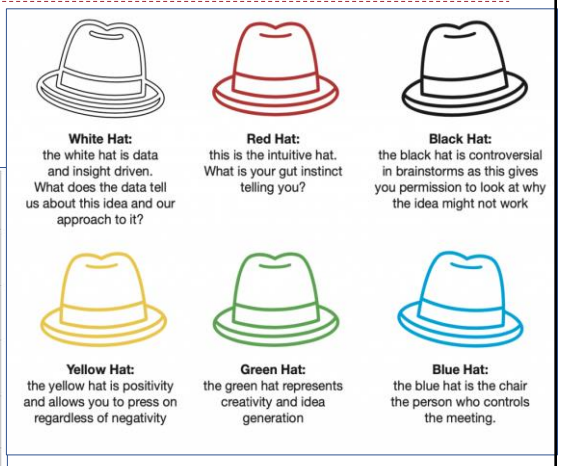
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Ideate

- Generate Ideas
  - Bono Thinking Hats

White Hat Facts		Facts - Names, Numbers, Indisputable
Red Hat Emotions		Emotions - I feel.....
Black Hat Negatives		The danger is that .....
Yellow Hat Positive		Positive - The benefits will be .....
Green Hat Creativity		If only ..... I would love .....
Blue Hat Overview		Recap, Summary of Next Steps



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Ideate

- Example (6 thinking hat)
  - Facilitate the conversation (wearing the blue hat).
  - White Hat: Present the facts of the problem
  - Green Hat: Generate ideas on how the problem can be solved
  - Yellow Hat: Evaluate the ideas by listing their benefits
  - Black Hat: Evaluate the ideas by listing their drawbacks
  - Red Hat: Get everybody's gut feelings about the alternatives
  - Blue Hat: Summaries the discussion and agree on the conclusions

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Ideate

- Generate Ideas
  - Plus 5
    - Generate 5 more ideas based on a colleague idea.
  - Place into client (or competitor) shoes

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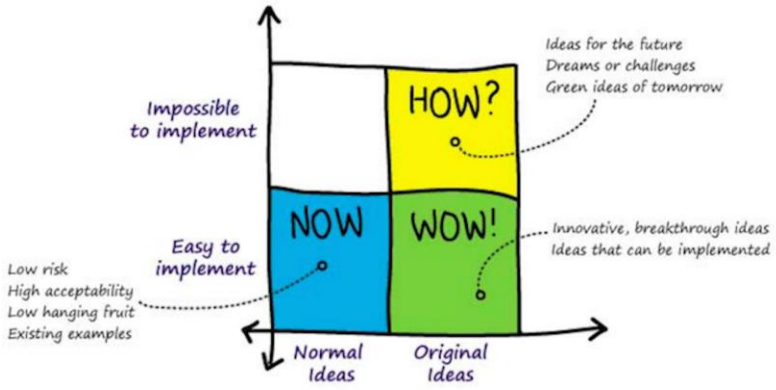
Ideate - Select and Sort Ideas

- Cluster
  - Organize ideas by topic
- Dot Voting
  - Each person has a number of dots to elect the best ideas.
- Idea Filter (Disney Method)
- 2x2 Matrix (How-Now-Wow)

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Ideate - Select and Sort Ideas

- 2x2 Matrix (Now-Wow-How)



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Prototype

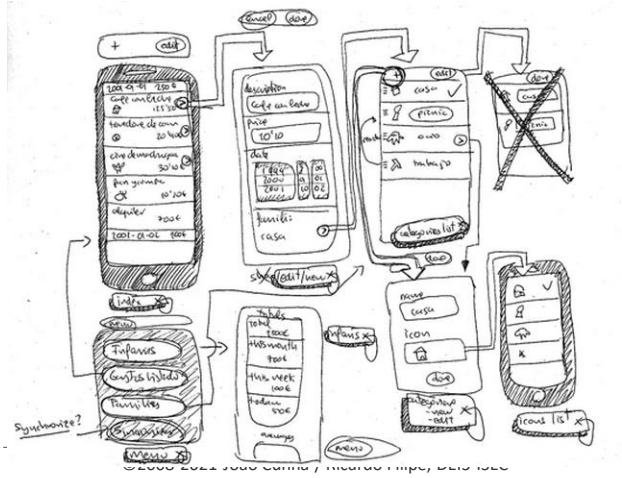
- Develop a prototype (concept) based on the best idea of ideation.
- Fast implementation prototype.
- Distinct techniques.

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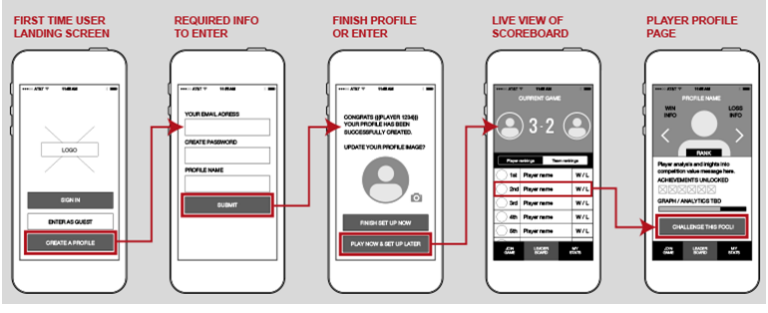
Prototype

- Distinct techniques:
  - Wireframes



Prototype

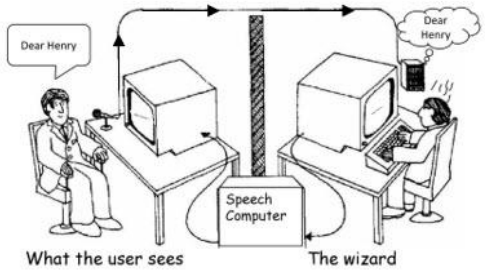
- Distinct techniques:
  - Click Dummies (figma is a good tool for this)



Prototype

- Distinct techniques:
  - Wizard of Oz
    - User thinks it is interacting with a computer, but a developer is responding to output.
    - Done in early stages to understand user expectations.

Wizard of Oz testing – The listening type writer IBM 1984

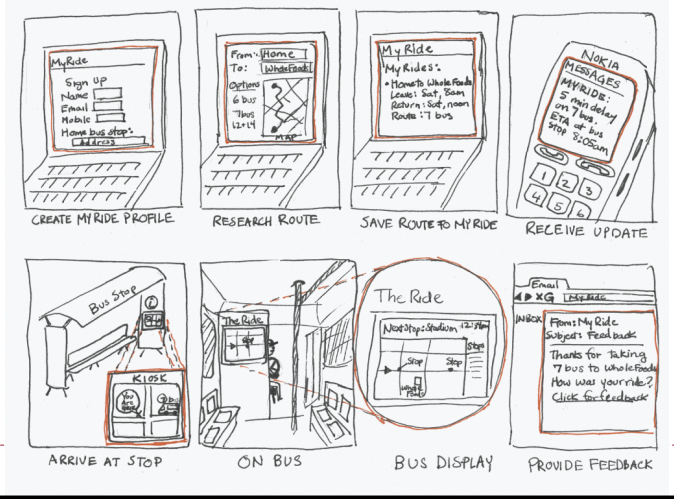


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Prototype

- Distinct techniques:
- Storyboards



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Prototype

- Distinct techniques:
- Lego (build something)



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Prototype

- https://youtu.be/jTageuhPfAM

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Test

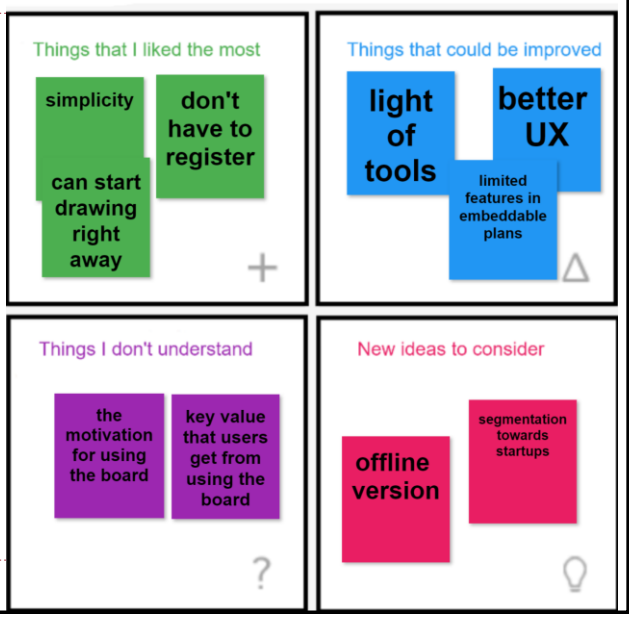
- Show your prototype to a testing audience.
- Be empathic
- Focus on hypothesis and goal
- Accept failure as part of the process.
- Need to watch how their customer uses and misuses the prototype
- Rollback to previous phases if needed (feedback from testing phase is essential).
- Team should take note of what their customer liked and didn't like, what questions and ideas emerged.

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Test

- Collect feedback

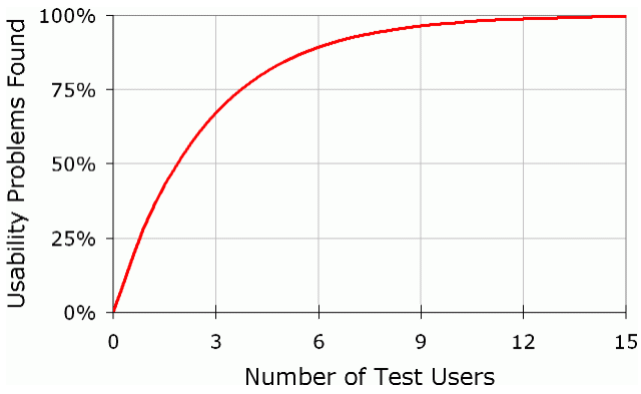


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Test

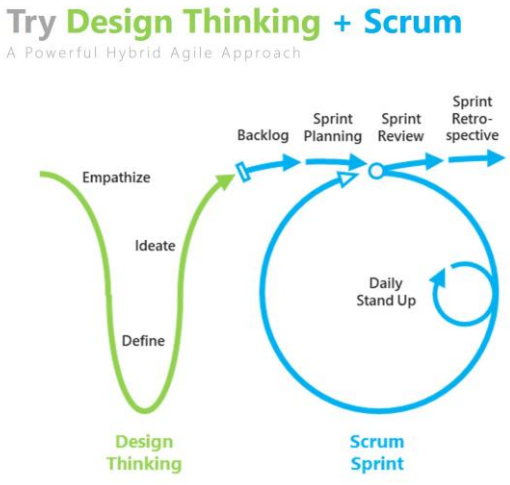
- Usability tests with 10-15 people max.



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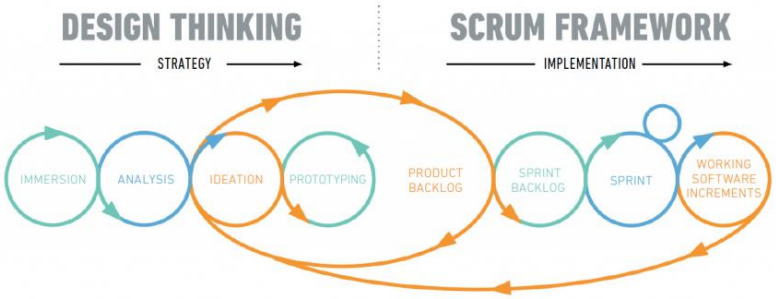
Design Thinking + Scrum (good combination?)



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Design Thinking + Scrum



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Bibliography

- An Introduction to Design Thinking PROCESS GUIDE
  - https://web.stanford.edu/~mshanks/MichaelShanks/files/509554.pdf
- https://subscription.packtpub.com/book/web-development/9781838645649/16/ch16lvl1sec78/design-thinking
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- https://www.margotlieblich.com/stanford-wallet-project
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